Orbiting Minds

An RR-10 Data-to-3D Simulation Experience

From spaceflight mouse-brain transcriptomics to interactive visual evidence and creative media.

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How can open space-biology data become more interpretable, inspectable, and memorable?

01 // BRIDGING THE ANALYTICAL GAP

By bridging the gap between raw datasets and interactive media, this project demonstrates how open transcriptomics data can be communicated beyond static plots.

02 // HYBRID EVIDENCE INTERPRETATION

While a benchmark tests how AI systems interpret structured scientific evidence, an interactive visual layer keeps the original evidence visible, creating a more accessible narrative.

> **PROJECT LIMITATION //** *Ultimately, visualization extends analysis communication; it does not replace primary analysis.*

RR-10: Spaceflight Mouse-Brain Transcriptomics

Mission Context

NASA Rodent Research-10

The NASA Rodent Research-10 (RR-10) mission provides the baseline scientific context for comparing spaceflight-induced physiological changes against earthbound baselines.

This comparative analysis strictly evaluates Flight (FLT) and Ground Control (GC) mouse samples, mapping region-specific neurological adaptation.

Focus is localized to two distinct functional brain regions: the **cerebellum** and the **hippocampus**, tracking transcriptional variations and sequence provenance.

OSDR Source Study / Region	Provenance ID
OSD-563 Cerebellum Transcriptomics	**GLDS-568** Dataset Registry
OSD-564 Hippocampus Transcriptomics	**GLDS-569** Dataset Registry
Analytical Conditions FLT (Flight Samples) vs. GC (Ground Control Baseline)	Organism **Mus musculus** RR-10 Cohort

REGION 01 // CEREBELLAR TISSUE

Cerebellum Comparison

An investigation of cerebellar architecture, evaluating spatial variance and transcriptomic stability between flight conditions and terrestrial baselines.

FLT PROFILE

Derived transcriptomic parameters show elevated dispersion in outer cerebellar layers.

GC PROFILE

Ground control baseline shows uniform architectural stability under standard conditions.

REGION 02 // HIPPOCAMPAL TISSUE

Hippocampus Comparison

Evaluating hippocampal tissue profiles to analyze how microgravity-induced stresses alter neural synaptic transcript mappings compared to ground groups.

FLT PROFILE

Pronounced contrast in synaptic transmission transcripts across hippocampal CA1/CA3.

GC PROFILE

Ground control maintains expected spatial patterns of localized genetic expression.

> *Visualization extends analysis communication; it does not replace primary analysis.*

From source records to reproducible notebook evidence

The spaceflight transcriptomics pipeline translates raw NASA OSDR source metadata into analytical benchmarks, generating structured datasets and procedural 3D visual environments.

STEP 01
NASA OSDR
Source records (OSD-563, OSD-564)

STEP 02
Kaggle DB
Translated public dataset

STEP 03
Notebook
Cerebellum vs Hippocampus analysis

STEP 04
Benchmark
RR-10 transcriptomics reasoning

STEP 05
JSON Truth
Processed biological data matrix

STEP 06
Blender
Procedural tissue & flight visualization

STEP 07
GLB / WebGL
Interactive model experience

FLT (FLIGHT) GC (GROUND)

Transcriptomic Space Analysis

Distinct PCA Space Positions

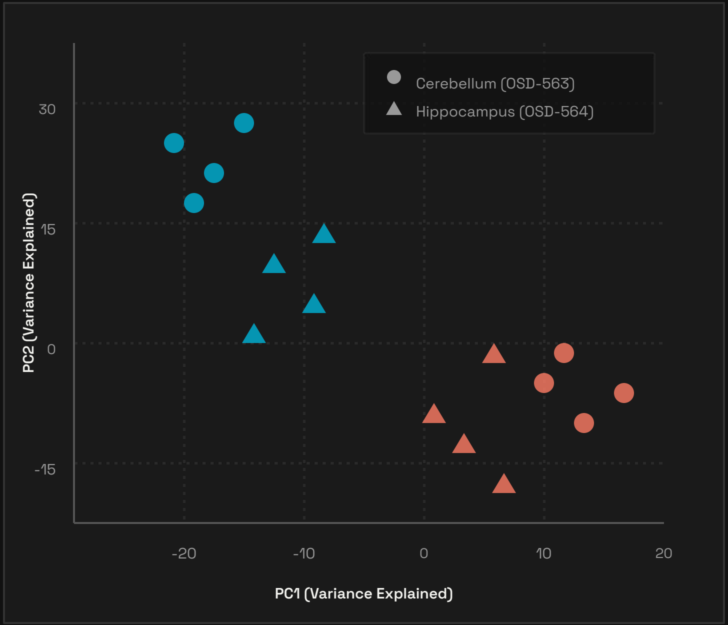
Flight (**FLT**) and Ground Control (**GC**) samples occupy distinct positions in PCA space. This Principal Component Analysis provides clear visual evidence of the variance between the sample conditions.

Cerebellum (OSD-563)

PC1 accounts for **25.6%** and PC2 for **18.5%**.

Hippocampus (OSD-564)

PC1 represents **28.3%** and PC2 is **15.7%**.



FLT (FLIGHT) GC (GROUND)

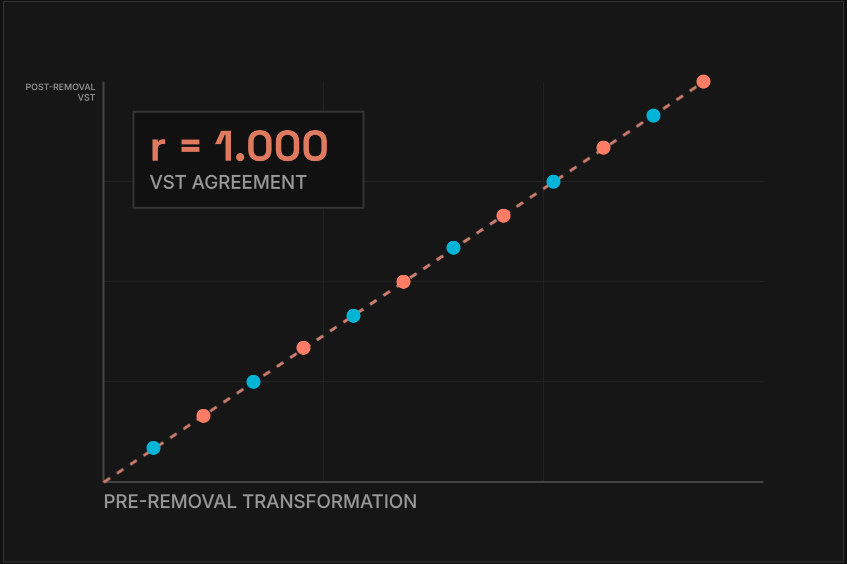


FIG 01 // VST AGREEMENT PLOT

Identity line y = x makes absolute value-stabilizing transformation agreement visible.

DATA INTEGRITY & BENCHMARKING

Quality-control context before downstream interpretation

The rRNA-removal quality control figure provides essential processing context rather than a biological outcome.

Per-value variance-stabilizing transformation (VST) agreement is displayed as **r = 1.000**, with the identity line $y = x$ making this agreement clearly visible.

Additionally, the distribution panel documents the rRNA-removal shift, ensuring data integrity before downstream interpretation.

BENCHMARK GOAL

Turning analysis context into an AI reasoning benchmark

This benchmark tests how AI models interpret dense scientific and transcriptomic context. The evaluation task is strictly grounded in the dataset, the analysis notebook, and an explicit evaluation design.

The project is specifically engineered to surface AI reasoning quality, evidence use, limitations, and uncertainty, rather than just generating a polished answer.

CRITERIA 01 // GROUNDED CONTEXT

Evaluation task is strictly tied to the provided biological dataset, corresponding analysis notebook, and physical constraints.

CRITERIA 02 // REASONING QUALITY

Assesses how models parse dense technical evidence, synthesize complex transcriptomic relationships, and formulate explicit logic.

CRITERIA 03 // UNCERTAINTY & LIMITS

Measures the model's ability to outline transcriptomic dataset limitations, establish boundaries, and cleanly report technical uncertainty.

The technical scene is driven by defined visualization mappings

Input Evidence Context	3D Visual Encoding	Interpretation Boundary
Derived adaptation & performance-like parameters	Neural-core scale and emission	Visualization control, not physical metric
Volatility-like parameters	Rhythmic pulse and asymmetry	Visualization control, not physical metric
Cluster and complexity parameters	Telemetry ring and probe density	Visualization control, not physical metric
Duration parameters & PCA coordinates	Animation timeline & constellation motif	Visualization control, not physical metric

Interpretation Warning These are derived visualization controls, **not direct physiological measurements.**

SYSTEM VIEWPORT // NEURAL PROXY

A data-synchronized murine neural proxy

The multi-mode WebGL interactive technical viewer delivers a real-time, data-synchronized visualization of murine brain region transcriptomics.

This neural proxy environment explicitly explains the connection between the interactive visual outputs and the underlying OSDR scientific records.

By maintaining rigorous boundaries, the viewer ensures audiences understand exactly how the source data informs the 3D experience without confusing it for a literal biological model.

SYS.LOC: OSDR-MP-09 // LATERAL_PROJ

DATA_MATRIX_SYNCHRONIZED

■ FLT_CYAN // ■ GC_CORAL // ■ BIO_CLAY

RENDER_ENGINE: WebGL2 // 60 FPS

Multi-mode interactive inspection of the neural proxy

01 // PRESENTATION

Offers a clean, cinematic scene tailored for high-level technical storytelling and clear overview.

02 // ANNOTATED

Highlights selected anatomical and telemetry hotspots with context to connect structural locations to real-time metrics.

03 // DIAGNOSTIC

Allows for detailed, procedural wireframe and topology inspection of the underlying spatial framework.

EVIDENCE DRAWER

Analytic Verification

Ensures literal PCA and QC graphics remain available for direct verification inside the interactive workspace.

** Note: The PCA constellation is graph-derived and does not represent a spatial anatomy map.*

Technical evidence and creative communication can reinforce each other

01 // ANALYTICAL VIEWPORT

Technical Benchmark

A rigorous, data-linked repository designed for scientific validation and reproducibility.

- Reproducible Pipeline: Built on a standardized, open-source workflow.
- Evidence Drawer: Interactive, fully structured data tables.
- Strict Boundaries: Explicit annotations of analytical limits.

02 // NARRATIVE VIEWPORT

Media Companion

A premium, human-readable editorial layer crafted for multidisciplinary engagement.

- Character-Led Journey: Guided onboarding and storytelling.
- Premium Visual Identity: High-end editorial layouts and graphics.
- Creative Narratives: Clearly labeled, contextual explanations.

> “One shared project architecture; two distinct modes of understanding.”
>
> Together, they bridge the gap between rigorous, complex biological data and intuitive, accessible communication.

NOT SCIENTIFIC EVIDENCE

Commander Pip: Zero-G Odyssey — Cinematic Media Companion

CHARACTER & HABITAT ARCHITECTURE

A stylized astronaut-mouse narrative inside a zero-G orbital habitat, modeled with real-time PBR shaders and web-optimized geometry in Blender.

- **EVA Spacesuit:** Matte-white insulated fabric, 4-point orange cross-harness
- **Bubble Visor:** Ultra-wide optical glass with warm-gold reflective sheen
- **Attitude Stabilizer:** 5-segment articulated tail with T-wing micro-thruster
- **Habitat Staging:** Zero-G floating nutrient orbs & Earth-horizon orbital glow

This is a stylized narrative experience inspired by the RR-10 mission context, separate from technical benchmark visualizations to provide public storytelling.

Citation, Responsible Reuse, & Resources

01 // NASA OSDR / RR-10

NASA OSDR RR-10 Studies (OSD-563 / OSD-564) original repository records.

osdr.nasa.gov

02 // KAGGLE DATASETS

Processed space-flight transcriptomic datasets & analytics notebooks.

kaggle.com/benchmarks/.../rr-10-benchmark

kaggle.com/code/.../rr-10-spaceflight-mice

03 // COMMUNITY & DEV

Interactive benchmark proxy, code repository, and deployment.

orbiting-minds.vercel.app

github.com/.../blender-simulator

PROJECT CITATION

Gaston Dana. ”Orbiting Minds” (GitHub, 2026)

Please cite this resource alongside the original NASA OSDR/GeneLab repository records for downstream applications and analytical replication.

Presenter Contact: Gaston Dana via GitHub • Google Scholar

RESPONSIBLE REUSE & LIMITATIONS

- **Interactive Proxy:** The technical visualization serves as a benchmark-informed interactive proxy, not a validated biomedical simulator.
- **Authority:** Original NASA OSDR records remain the authoritative source.
- **Media Companion:** All narrative companion materials represent stylized creative content, not scientific evidence.